

RebelCast: Forecasting rebel organization behavior

Nils W. Metternich
CONflict Analysis LAB (COALA)
presentation includes contributions by:
Gokhan Ciflikli (LSE)
Altaf Ali (UCL)
Elio Amicarelli (UCL)

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Department of Political Science
University College London
n.metternich@ucl.ac.uk

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- ▶ Going beyond structural factors to explain onset, duration, escalation, and recurrence
 - ▶ Demand to go beyond structural factors
 - ▶ Focus on actors that actually make decisions about conflict
- ▶ Predicting conflict dynamics (escalation, duration)
 - ▶ Machine Learning approaches help to improve sensitivity-specificity tradeoff (true-positives/true-negatives) (Muchlinski et al. 2016)
 - ▶ Ensemble Bayesian Model Averaging models help us to integrate the predictions from several models (Montgomery, Hollenbach, and Ward, 2012)

Hollenbach et al. (2017) on imputation

Predictive turn in conflict studies

- ▶ Gurr and Lichbach, 1986
- ▶ O'brien, 2002
- ▶ Schrodtt, 2006
- ▶ Goldstone et al., 2010
- ▶ Ward, 2010
- ▶ Weidmann and Ward, 2010
- ▶ Schneider, Gleditsch and Carey, 2011
- ▶ De Mesquita, 2011
- ▶ Ward et al., 2013
- ▶ Gleditsch and Ward, 2013
- ▶ Hegre et al., 2013
- ▶ Bell et al., 2013
- ▶ Brandt, Freeman and Schrodtt, 2014
- ▶ Muchlinski et al., 2016

Characteristics of predictive approaches

- ▶ $Y = f(X) + \epsilon$
- ▶ Focus on $\hat{Y}$ rather than $\hat{\beta}$
- ▶ $\hat{Y} = \hat{f}$

- ▶ Based on the insights of interdependent actors, we are developing two forecasting models
 - ▶ RebelCast Escalation
 - Country
 - Rebel
 - ▶ RebelCast Duration
 - Country
 - Rebel
- ▶ Both use data from RebelTrack
 - ▶ R package calculating monthly rebel organization measures based on UCDP GED
 - Area active
 - Event count
 - Distance to capital
 - Mean fighting position
 - Distance to other rebel organizations
 - etc

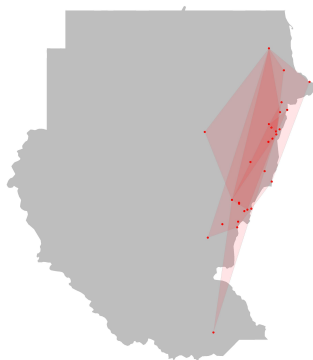


Figure: NDA (National Democratic Alliance). Hull area of fighting activity

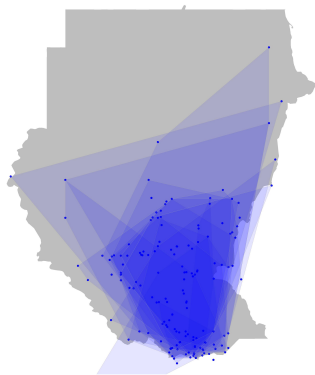


Figure: SPLM/A (South People's Liberation Movement). Hull area of fighting activity

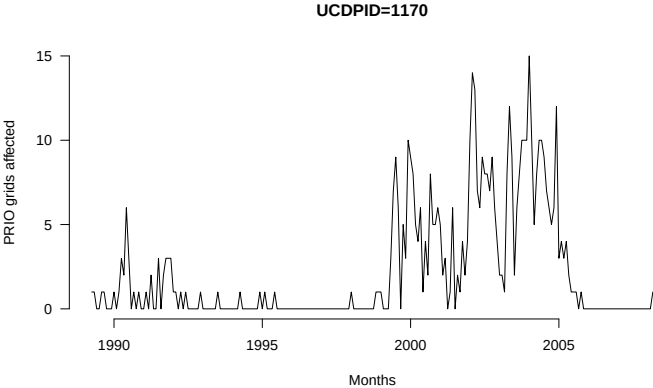


Figure: GAM (Free Aceh Movement). Number of active PRIO grids (0.5×0.5 decimal degree) per month

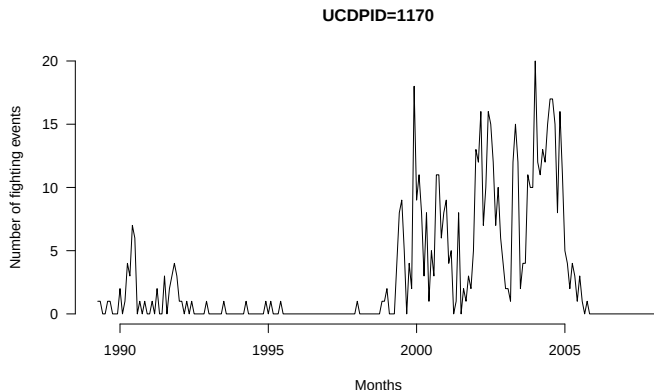


Figure: GAM (Free Aceh Movement). Number of active days per month

RebelCast Duration

When does the fighting stop?

Machine learning approach

- ▶ EBMA
- ▶ Outcome: UCDP GED Rebel spells (Fighting duration)
- ▶ Input
 - ▶ RebelCast variables
 - ▶ ICEWS/Phoenix

Explanatory approaches to civil conflict duration

► Structural

- Poverty (e.g. Collier, Hoeffler and Soderbom, 2004)
- Low state capacity (e.g. Mason and Fett, 1996)
- Resources (e.g. Buhaug, Gates and Lujala, 2009)
- Sons of Soil (e.g. Fearon, 2004)

► Third parties

- (e.g. Regan, 2002)

► Rebel organization characteristics

- Strength (e.g. Cunningham, Gleditsch and Salehyan, 2009)
- Tactics (e.g. Balcells and Kalyvas, 2014)
- Ethnic Linkage (e.g. Wucherpfennig, Metternich, Cederman, and Gleditsch, 2012)

► Rebel organization configurations

- Veto (e.g. Cunningham, 2006)
- Alliances (e.g. Christia, 2012)

Predictive approaches to civil conflict duration

► Predictive

- Pilster and Böhmelt, 2014
- Chiba, Metternich, Ward, 2015
- Mason, Weingarten, and Fett, 1999

Predictive approaches to conflict duration

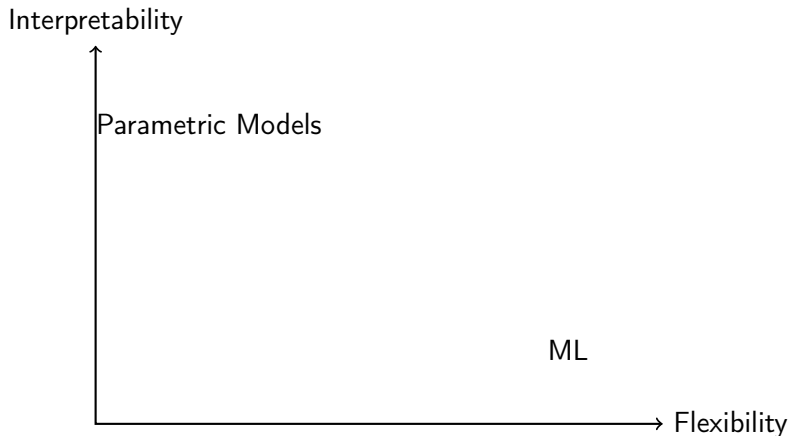


Table: Cox Proportional Hazards Estimates, Wucherpfennig et al. (2012)

	<i>Dependent variable:</i>	
	<i>War Duration</i>	
	<i>Cox Prop. Hazards TVC[†]</i>	<i>Cox Prop. Hazards Non-TVC[‡]</i>
Ethnic Linkage with Included Group	0.307 (0.234)	0.374 (0.145)
Ethnic Linkage with Excluded Group	-0.419** (0.165)	-0.415 (0.066)
Territorial Conflict	0.061 (0.172)	0.113 (0.068)
Strong Central Command	0.453*** (0.151)	0.132 (0.078)
Legal Political Wing	0.356* (0.166)	0.033 (0.084)
Territorial Control	-0.350** (0.150)	-1.046*** (0.067)
Democracy	-0.849*** (0.174)	-1.274*** (0.066)
ln GDP p.c.	0.086 (0.074)	-0.008 (0.031)
ln Population	-0.044 (0.049)	0.049 (0.021)
Natural Resources	-0.354** (0.143)	-0.730*** (0.064)
Observations	1,941	1,921
Log Likelihood	-1,143.255	-10,031.730
Wald Test (df = 10)	73.340***	72.270***
LR Test (df = 10)	71.003***	918.340***
Score (Logrank) Test (df = 10)	67.764***	927.270***

Non-clustered standard errors in parentheses

*p-values for clustered s.e. *p<0.1; **p<0.05; ***p<0.01*

[†] Time-varying covariates

[‡] Variables are fixed to their onset-year values

ML: Predicting the end of conflicts

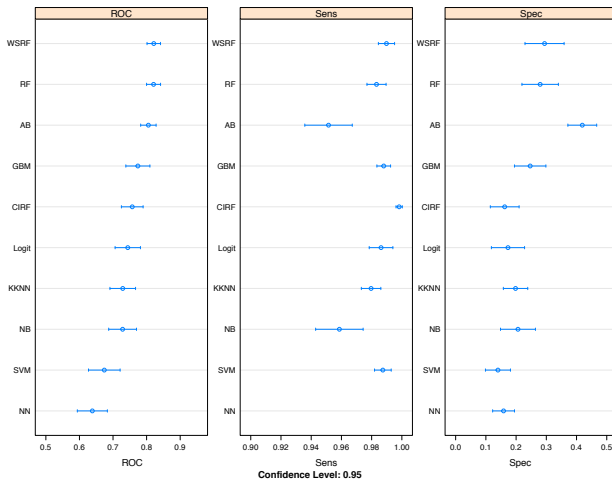


Figure: Machine learning predictions: Repeated 10k-fold Cross-Validation

EBMA: Predicting the end of conflicts

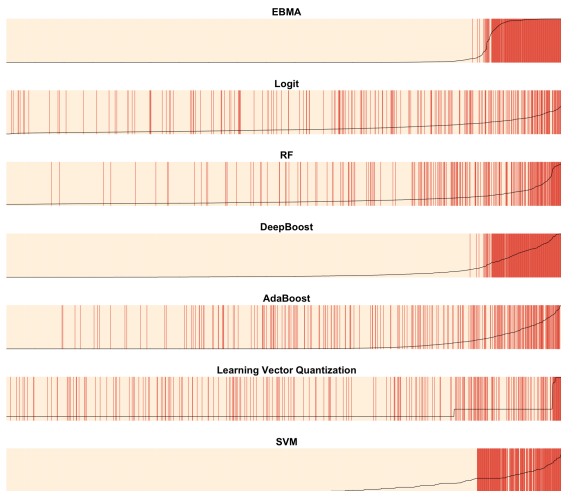


Figure: EBMA predictions: Repeated 10k-fold Cross-Validation

EBMA: Predicting the end of conflicts

	brier	auc	perCorrect
EBMA	0.008592138	0.9988799	0.9871200
Logit	0.100156477	0.7553253	0.8737764
RF	0.081079930	0.8689720	0.8902628
DeepBoost	0.029883143	0.9986915	0.9536321
AdaBoost	0.115938911	0.7698648	0.8598660
Learning Vector Quantization	0.097659379	0.6893999	0.8845956
SVM	0.050751267	0.9885928	0.9103555

Figure: Performance statistics: Brier score: average squared deviation of the predicted probability. AUC: Area under the Receiver-Operating Characteristic curve. perCorrect: percentage correctly predicted (.5 cut-off)

Conclusion and looking ahead

- ▶ ML approaches allow for better Early Warning models
- ▶ EBMA allows to weigh strengths of different models